

MergeNet: Spatial Routing Gains and Deployment Tradeoffs in Learned Token Compression

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Abstract

Spatial constraints can improve learned token aggregation without making the resulting model an efficient alternative to dense computation. We examine this distinction in MergeNet, a DTEM-derived classifier that performs original-grid-constrained soft mass transfer, selects a fixed number of carriers, and recovers information before latent processing. In a single-seed ImageNet-1k campaign, radius-3 routing improves 150-epoch top-1 accuracy by 1.670 percentage points over an unconstrained router and 0.598 points over a flattened-window control. An independent CIFAR-100 campaign with three seeds, two selectors, and a degree-matched support control reproduces a radius-3 advantage in every paired comparison, including $+1.42 \pm 0.30$ points over receiver-permuted degree-matched support under the historical selector. The full ImageNet model reaches 81.368% after 300 epochs at 224 pixels, versus 82.246% for dense DeiT-S/8. Fine-tuning narrows the gap to 82.582% versus 82.976% at 384 pixels, then 82.618% versus 83.068% at 512 pixels: the dense-relative deficit does not keep closing with resolution. Four completed 150-epoch probes, including longer warmup, a λ curriculum, a higher learning rate, and matched stochastic depth, remain inside a campaign-defined ± 0.40 point decision band around the selected configuration. Despite 36.9% fewer estimated multiply-accumulates, MergeNet takes 7.1% more inference time in a guarded 224-pixel synthetic H2O benchmark. A 384-pixel batch-64 radius-3 benchmark reduces inference time by 20.0% relative to dense DeiT, while an independent A100 harness finds no dense-relative speedup at either resolution on the local stack. Post-training ToMe has higher recorded accuracy at all twenty matched final-token budgets. The study separates internal routing gains from deployment evidence and identifies the remaining representation-level checks that still require collaborator-side checkpoints.

Author working draft with requested evidence slots P01–P12.

Underlined cells and reserved figures are intentionally unfilled; they are not observations and do not enter any reported result.

1 Introduction

Fine patch grids preserve spatial detail but increase the cost of vision transformers. Token merging offers a way to retain information from that grid while reducing the sequence processed by later layers [1]. Learning the merge criterion can make compression responsive to the task, but a useful learned router must satisfy two distinct requirements: its aggregation decisions must preserve useful representations, and its execution must save enough work to pay for routing itself. An improvement in the first requirement does not imply an improvement in the second.

We study this distinction through MergeNet, a classifier that combines a DTEM-derived soft router [6] with a full-grid local encoder, a fixed carrier budget, global recovery attention, and a compressed latent encoder. The routing constraint is simple: a donor can send mass only to receiver slots within a Euclidean radius on the original image grid. The constraint prevents a flattened sequence from treating a row boundary as ordinary spatial adjacency. It also limits candidate support independently of the learned similarity. We compare this router with unconstrained, flattened-window, and degree-matched versions of the same architecture before comparing the complete model with dense and post-training compression baselines.

The resulting evidence supports a qualified conclusion. With matched 150-epoch ImageNet settings, radius-3 routing improves best validation accuracy by 1.670 *pp* over the unconstrained router and 0.598 *pp* over the flattened window. These are observed single-seed improvements. They do not, by themselves, separate geometry from candidate cardinality. A later CIFAR-100 campaign with three seeds and a degree-matched receiver permutation of the realized radius-3 mask finds a radius-3 advantage in all paired comparisons, under both the historical batch-normalized selector and a per-sample bisection selector.

The deployment evidence is less favorable. At 224 pixels, a 300-epoch MergeNet reaches 81.368% top-1, compared with 82.246% for dense DeiT-S/8. Its batch-64 inference takes 45.71 ms on a guarded H20 benchmark, compared with 42.68 ms for the dense model, despite lower estimated arithmetic cost. After 384-pixel fine-tuning, the accuracy gap narrows to 0.394 *pp*; a further 512-pixel stage leaves a 0.450 *pp* gap. Completed optimizer-side probes do not move the selected 150-epoch configuration outside a campaign-defined 0.40 *pp* band. Post-training ToMe remains a strong accuracy baseline at matched final patch counts. The available ToMe accuracy and latency harnesses use different proportional-attention settings, so their outputs cannot be combined into a verified accuracy-latency frontier.

Our contributions are an implementation-specific analysis of spatially constrained soft routing, a traceable evaluation of its internal and external comparisons, including a three-seed CIFAR confirmation and a completed resolution series, and a diagnosis of why nominal compression and realized savings can diverge. We explicitly distinguish the support of a routing step from the receptive field of a feature, and soft mass transfer from physical sequence reduction. The study establishes useful internal controls and concrete reproduction requirements; it does not claim state-of-the-art compression, priority for spatial merging, or a general failure of learned token merging.

2 Related Work

Token reduction methods differ in which tokens they retain, how they aggregate information, and where they reduce computation. DynamicViT learns token retention decisions and trains them using differentiable attention masking [10], while A-ViT assigns adaptive computation depth through token halting [17]. Token Merging (ToMe) combines similar tokens through bipartite soft matching and can be applied to an existing checkpoint without additional training or incorporated into training [1]. Subsequent work changes the merge criterion: PiToMe protects informative or isolated tokens using an energy score [13]; MCTF combines similarity, informativeness, and token size with one-step-ahead attention and reduction-consistency training [7]; SAD-TM uses visual saliency and a deferred-merging variant [15]. These methods provide stronger comparisons for a compression mechanism than an uncompressed backbone alone.

The closest methodological predecessor of MergeNet is Decoupled Token Embedding for Merging (DTEM), which learns low-dimensional merge embeddings through continuously relaxed grouping and merging [6]. DTEM separates the representation used to select merges from the features used by the backbone. Its soft embedding updates retain the physical token count; end-to-end training alternates these updates with backbone updates that use hard merging, and inference uses bipartite matching. MergeNet builds on this lineage through the OpenToMe implementation [14]. Our study concerns the support of the routing operator: which donor-receiver pairs are eligible under a prescribed carrier budget. The decoupled embedding and soft merge operators are inherited components. The unconstrained MergeNet variant is an internal architectural control and should not be interpreted as a reproduction of the original DTEM training and inference protocol.

Spatial locality is also an established design choice for token reduction. DSM delays merging, applies it within spatial windows, and then transitions to global merging [4]. ALGM similarly combines early local merging with later global merging for semantic segmentation [9]. GTP-ViT propagates tokens over a graph that combines adjacency on the original image grid with semantic connections [16]. More recent spatial reducers impose different constraints: CubistMerge restricts matching to neighbors along rows and columns while preserving a rectangular token layout [3], whereas MAT merges adjacent tokens after Hilbert-curve reordering [8]. MergeNet instead defines eligible routes by a Euclidean radius on the original patch grid. This distinction motivates controlled comparisons of global, flattened-window, and two-dimensional routing support; it does not establish priority for locality or for hard spatial constraints.

Compression and recovery must also be distinguished from the routing rule. LookupViT processes a fixed set of compressed tokens alongside a cheaper high-resolution token stream, exchanging information through bidirectional cross-attention [5]. ALGM restores token resolution for dense prediction [9], while DToP uses early predictions for confident segmentation tokens and combines predictions across stages without reconstructing those tokens [11]. These methods show that retaining or recovering dense information is a separate architectural choice. Our classification study evaluates the behavior and cost of a particular local-latent architecture; it does not establish a new general recovery mechanism or its suitability for dense prediction.

Evaluation protocols determine what a comparison can establish. ToMe includes DeiT models trained from scratch for 300 epochs, whereas DTEM primarily studies adaptation of pretrained models [1, 6]. Published results also depend on the backbone and its training recipe [12]; they cannot substitute for controlled comparisons in our patch-8 setting. DiffRate further demonstrates that layerwise compression

rates are a design variable distinct from the merge criterion [2]. Equal final token counts therefore do not imply equal computation throughout the network. Moreover, DSM reports that data movement associated with local merging can limit throughput gains despite arithmetic savings [4]. We accordingly separate the internal geometry study from comparisons with post-training ToMe and PiToMe, and distinguish achieved token counts, arithmetic cost, and measured latency. The resulting scope is a study of spatial routing and its execution costs, with conclusions bounded by the evaluated models, training protocols, and hardware.

3 Spatially Constrained Soft Routing

3.1 Architecture and actual sequence reduction

For an image of size $H \times W$ and patch size p , let $N = (H/p)(W/p)$ denote the number of patch slots, excluding a protected class token. The main configuration uses $p = 8$, width $d = 384$, six attention heads, six local blocks, and six latent blocks. At 224×224 , the local encoder processes $N + 1 = 785$ tokens in every block. Its attention window is a one-dimensional band extending 16 sequence positions in each direction, with global attention for the class query. It is not a 16×16 spatial window. Two-dimensional geometry enters through the routing support below.

The local encoder first computes all six layer outputs, $H^{(1)}, \dots, H^{(L)}$. Routing then starts from the last output, $X^{(0)} = H^{(L)}$, using a separate metric derived from each $H^{(\ell)}$. Each routing step retains the complete slot tensor. Only after these steps are completed does a hard gather physically reduce the sequence. This order is essential: earlier local blocks do not benefit from progressively shorter sequences. Figure 1 depicts the executed dataflow.

3.2 Original-grid eligibility

Index patch slots by $i \in \{0, \dots, N - 1\}$ and define original coordinates

$$u_i = (\lfloor i/(W/p) \rfloor, i \bmod (W/p)). \quad (1)$$

At step ℓ , patches are partitioned into donor and receiver sets A_ℓ and B_ℓ . Training uses a random partition for each sample; evaluation alternates deterministic odd/even partitions across steps. The class token belongs to neither merging set. For donor i , eligible receivers are

$$\mathcal{E}_i^{(\ell)}(R) = \{j \in B_\ell : \|u_i - u_j\|_2 \leq R\}. \quad (2)$$

Coordinates refer to the original slots throughout routing. They are not replaced by the centers of the accumulated feature content. The main choice is $R = 3$ patch spacings. The global control removes this spatial constraint; the flattened control uses the release’s one-dimensional matching window with width parameter 8.

Let $z_i^{(\ell)}$ be the normalized output of a learned $384 \rightarrow 64$ metric projection applied to $H_i^{(\ell)}$. Its input gradient is scaled by 0.1 without changing its forward value. Following the inherited relaxed-routing construction, eligible similarities define

$$s_{ij} = \langle z_i^{(\ell)}, z_j^{(\ell)} \rangle / \tau_1, \quad (3)$$

$$e_i = \sum_{j \in \mathcal{E}_i^{(\ell)}} \exp(s_{ij}), \quad q_{ij} = \frac{\exp(s_{ij})}{e_i} \quad (j \in \mathcal{E}_i^{(\ell)}). \quad (4)$$

A threshold-based soft top- k operator converts row energies e_i into donor weights g_i , with the step’s scheduled merge budget as its target. We use $\tau_1 = 1$ and multiplicative soft-selection scale 30. The released threshold routine uses a batch-wide range normalization (Section 6.2); its requested budget does not guarantee exact relaxed selection mass for every numerical input. The assignment is $a_{ij} = g_i q_{ij}$ on eligible edges and zero elsewhere, with row-sum stabilization in the implementation. Empty rows have zero assignment. If fewer eligible donor rows exist than the requested budget, the spatial path raises an error instead of routing outside the support. Our notation suppresses the implementation’s numerical clamps and straight-through stabilizers; it does not assume the relaxed weights are exact binary choices.

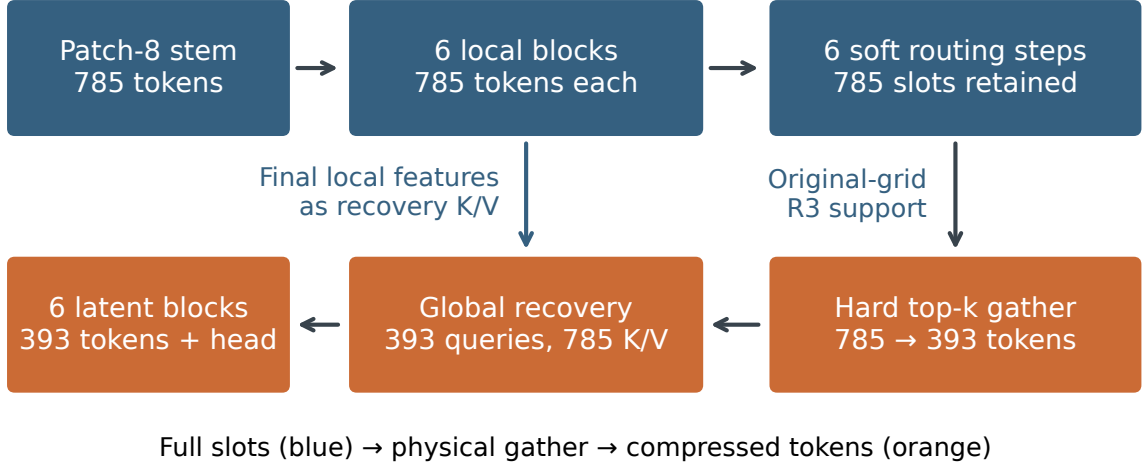


Figure 1: Main MergeNet dataflow at 224 pixels. All six local blocks run before routing; every soft routing step retains 785 slots. The hard gather is the first physical reduction. Recovery reads the final local features before soft transport. Blue marks full-grid processing and orange marks compressed processing. Numbers include the class token.

3.3 Mass transfer and fixed-budget selection

Each slot carries a nonnegative mass m_i , initialized to one. For a donor i and a receiver j , soft transport updates masses and receiver features as

$$m'_i = m_i \left(1 - \sum_j a_{ij} \right), \quad X'_i = X_i, \quad (5)$$

$$m'_j = m_j + \sum_i a_{ij} m_i, \quad (6)$$

$$X'_j = \frac{m_j X_j + \sum_i a_{ij} m_i X_i}{m'_j}. \quad (7)$$

The code clamps residual donor fractions and receiver denominators for numerical stability. Donor slots remain in the tensor even when their mass is small. Thus soft transport and token deletion are different operations. The main spatial path evaluates a sparse neighbor list and uses indexed transport; a dense score-then-mask path is available as a reference.

For compression parameter λ , the final number of patch carriers is

$$K = N - \left\lfloor N \frac{\lambda - 1}{\lambda} \right\rfloor. \quad (8)$$

The model gathers the K largest-mass patches and prepends the protected class token. At $\lambda = 2$, $K = 392$ at 224 pixels and $K = 1152$ at 384 pixels. At $\lambda = 2$, the six soft donor-budget targets are $(65, 65, 65, 65, 65, 67)$ at 224 pixels and $(192, 192, 192, 192, 192, 192)$ at 384. The carrier count is fixed by configuration; the selected locations depend on the input. A soft top- K surrogate supplies straight-through gradients through the selected features via $1 + g - \text{stopgrad}(g)$; it does not differentiate every gather index. This multiplier is computed at evaluation too and equals one for finite values, whereas non-finite values do not cancel. During training an auxiliary head contribution is added to the main logits, with maximum weight 0.05; it is absent during evaluation. This is a logit contribution, not a separately reported auxiliary cross-entropy loss.

The default source-tracing mode maintains a detached mass-weighted center of the flattened indices. It uses these values to sort selected carriers; it does not retain the full source-membership matrix. Selected features Q then undergo residual recovery,

$$Y = Q + \text{CrossAttn}(Q, H^{(L)}, H^{(L)}). \quad (9)$$

Recovery is global in this mode. Six global latent blocks process $K + 1$ tokens, with no further merging. Latent attention retains proportional attention: it adds the logarithm of the selected token mass to key logits before the softmax. The class representation feeds the classification head.

3.4 What the geometry and complexity claims mean

Equation (2) bounds a single transport edge. It does not bound the spatial extent of all original constituents after multiple steps: a sequence of locally permitted transfers can travel farther. Moreover, $H^{(L)}$ already contains contextual information and Equation (9) reads globally. The constraint therefore concerns routing support, not the entire model’s receptive field or semantic segmentation boundaries.

The distinction between slots and carriers also determines cost. Local attention projections and MLPs still process $N + 1$ tokens per layer. Recovery has attention terms proportional to NKd , and latent self-attention has terms proportional to K^2d . For fixed λ , both retain quadratic terms in N . Sparse routing avoids a dense donor–receiver assignment during the forward pass, but initialization still constructs an $N \times N$ Boolean adjacency buffer. Neither a linear-memory claim for the complete implementation nor an end-to-end speedup follows from the sparse scorer alone.

Reserved method variants. The original method above remains the subject of the completed experiments. The initialized and progressive-latent variants in Sections 5.6 and 5.7 have separate pending endpoints. They do not retrospectively change the original dataflow or its interpretation.

3.5 Measured routing structure: reserved visualization

Evidence pending [P06]: Trace actual routing on trained ImageNet examples

The figure is reserved for measured outputs, not an architectural illustration.

Preselect validation image IDs and include typical and unsuccessful cases. For each layer export the donor/receiver partition, eligible edges, transport weights, selected carriers and token masses. To visualize original-source membership, enable a verified membership trace: the default flattened-index center cannot reconstruct it. Report the sampling rule and checkpoint identity. Local edges alone do not establish semantic-boundary preservation; that claim requires an annotation source and an explicit metric.

P06: images, routing edges, carriers, and a failure case
Reserved space; no measurements or images supplied.

Figure 2: Reserved trained-checkpoint routing visualization [P06]. No synthetic image or hand-drawn assignment is used as a model observation.

Result and interpretation:

4 Experimental Protocol

4.1 Data, optimization, and run identity

We analyze an ImageNet-1k classification campaign with seed 42. Evaluation uses the 50,000-image validation split and reports top-1 accuracy of the exponential moving average (EMA) model. The September 19 packet contains 23 runs with recorded epochs: 21 complete schedules, a 140/150 weight-decay probe stopped by a scheduled reboot, and a DeiT-initialized run interrupted at 105/150. Endpoints

in this manuscript are recomputed from each run’s `summary.csv` by header name. A September 21 audit verifies 3,365 contiguous summary rows across all 23 runs and matches 3,264 rows’ EMA loss, top-1 and top-5 to supplied logs. Logs are missing for 86 curriculum epochs (64–149) and all 15 MergeNet 512-pixel epochs. These 101 rows remain summary-only evidence. Three conflicting early rows in one 384-pixel run are resolved using a documented restart and a later complete matching trajectory; the original conflicting records are retained. Checkpoint-verification receipts are present, but the weights themselves were not transferred with the campaign packet. This audit verifies the recorded outcomes; it is not an independent checkpoint re-evaluation.

The main 224-pixel scratch arms train for 150 or 300 epochs, using AdamW, cosine learning-rate decay, five warmup epochs (twenty for the warmup probe), weight decay 0.05 unless noted, gradient clipping at 1.0, label smoothing 0.1, Mixup 0.8, CutMix 1.0, RandAugment, and random erasing probability 0.25. The base learning rate is 5×10^{-4} ; a separate MergeNet arm uses 7.5×10^{-4} , and one probe uses 10^{-3} . The higher-rate 7.5×10^{-4} arm supplies the 300-epoch MergeNet checkpoint. Therefore the final dense-versus-MergeNet comparison is a comparison of selected configurations, while the geometry comparison uses matched learning rates. No teacher distillation is enabled. Appendix A gives the recipe and auxiliary-logit schedule.

The effective global batch is 1,024 throughout the recorded campaign. The 384/512-pixel fine-tunes use four-step gradient accumulation: $32 \times 8 \times 4 = 1024$ for the dense 384-pixel run and $64 \times 4 \times 4 = 1024$ for the other fine-tunes. Each fine-tune epoch contains 5,004 per-rank microsteps and 1,251 optimizer steps. Counting microsteps as optimizer steps would incorrectly suggest a fourfold batch reduction. Different world sizes and microbatches still produce different execution and sample-grouping conditions; effective batch equality alone does not establish bitwise equivalence.

4.2 Resolution adaptation and checkpoint selection

Three 384-pixel runs fine-tune the corresponding 300-epoch EMA weights for 30 epochs, using learning rate 10^{-5} , minimum rate 10^{-6} , one warmup epoch, weight decay 10^{-8} , and no Mixup or CutMix. EMA decay changes from 0.99996 to 0.9996. The MergeNet arms either retain $R = 3$ or use $R = 3 \times 384/224 \approx 5.143$. The latter tests a particular proportional radius scaling; it is not a complete radius search. Bicubic validation preprocessing uses crop percentage 0.9 at 224 pixels and 1.0 for the fine-tunes. The 512-pixel dense result follows its 384-pixel checkpoint. The paired 15-epoch MergeNet 512-pixel run is initialized from the radius-5.143 384-pixel EMA weights and uses $R = 3 \times 512/224 \approx 6.857$. The radius-3 384-pixel packet has no extracted EMA file, so that checkpoint cannot seed a 512-pixel stage without a further extraction on the training machine.

We report both the best observed EMA top-1 and the final-epoch EMA top-1. Best checkpoints are selected on the same validation split used for reporting, so these scores are not independent held-out estimates of model selection. Epoch indices start at zero. All completed-run comparisons use actual summary headers rather than fixed column positions, because the summary schema changes with enabled diagnostics. Runs with a teardown error after a complete recorded schedule are distinguished from unfinished runs.

4.3 Post-training compression and timing

The accuracy sweep applies ToMe and PiToMe to the dense EMA checkpoint without additional training, and changes MergeNet’s carrier budget without retraining each point. Thus MergeNet has been trained with its native compression, whereas the dense backbone was trained without merging. This is a deployment comparison of available checkpoints, not an isolation of the routing rule under identical training. We distinguish a 224-pixel checkpoint evaluated at a new resolution from a checkpoint fine-tuned at that resolution. Repeated successful sweep records are resolved by latest timestamp, not highest accuracy. All included accuracy cells use the full validation split. Training-summary validation uses 64 images per rank. For sweeps from the 224-pixel checkpoints, MergeNet and ToMe use batch 128 and PiToMe uses batch 256; the final 384-pixel fine-tuned curves use batch 256 for all three. The separate fine-tuned gate uses batch 384. These batch distinctions matter for the soft-routing implementation and are retained in the evidence ledger.

The 224-pixel efficiency experiment uses one H20-3e, synthetic inputs, batch 64, fp16 autocast, 30 warmup iterations and 50 measured iterations. It records CUDA-event medians and interquartile ranges (IQRs); training cells include forward, loss, backward and AdamW update. Device isolation is checked around the measurements. The separate radius-3 benchmarks at 384 pixels use the same

thirty-warmup/fifty-measurement protocol at batches 64 and 128. These are model-execution timings, excluding image loading, and do not estimate end-to-end training time on eight GPUs. Training-log throughput is excluded because the training machine was shared.

A later 384-pixel benchmark uses batch 64, ten warmup and 30 measured iterations, random model initialization and synthetic inputs. It records achieved token counts, but disables ToMe proportional attention, whereas the accuracy sweep uses it. We retain these latency measurements as a separate configuration-specific experiment and do not attach accuracy coordinates to them. Final token counts also require care: the accuracy sweep counts patches, while this timing harness counts the class token as well.

4.4 Protocol for reserved extensions

The working-draft slots P01–P12 specify additional evidence without adding observations to the completed-run tables. Initialization and progressive latent experiments are defined in Sections 5.6 and 5.7; their resolved arguments and source-state receipts must be fixed before interpretation. ImageNet support replication uses paired training seeds (Section 5.2). Joint accuracy and latency require the configuration identity in Section 6.3. Additional metadata and currently missing log segments are enumerated in Appendix H.4.

5 ImageNet Results

5.1 Internal geometry controls

Table 1 separates routing controls from later optimization probes. A historical single-seed CIFAR-100 sweep is reported in Appendix F; a three-seed confirmation is in Section 5.3. At learning rate 5×10^{-4} , the radius-3 model reaches 78.888%, versus 77.218% for global routing and 78.290% for flattened-window routing. The corresponding improvements are 1.670 *pp* and 0.598 *pp*. Their final-epoch differences are 1.672 *pp* and 0.610 *pp*, so best-checkpoint selection does not explain the ordering. Radius 2 reaches 78.942%, only 0.054 *pp* above radius 3. With one seed, this does not establish an optimal radius or statistical equivalence between the two choices.

The geometric result is specific to these support policies. Radius changes alter both spatial coverage and the number of eligible receivers. Since row energies sum over eligible receiver scores, candidate cardinality can affect donor selection even before transport. The ImageNet packet has no degree-matched randomized support control. Likewise, global MergeNet is not the original DTEM method: it shares MergeNet’s local front end, soft inference path, final gather, and recovery operation.

Raising the learning rate to 7.5×10^{-4} improves radius-3 accuracy by 0.428 *pp* at 150 epochs. That selected configuration is the baseline for the optimizer probes in Section 5.5. A separate $\lambda = 4$ run reaches 78.366%, but also changes local/latent depths from 6/6 to 4/8 and the local window from 16 to 32; it cannot be interpreted as a compression-ratio-only experiment.

5.2 ImageNet geometry confirmation: reserved experiment

Evidence pending [P04]: Replicated ImageNet support controls

These cells are intentionally blank and are not included in the reported results.

The proposed extension compares global, flattened-window, radius-3 and degree-matched support with identical depth, learning rate, stochastic depth, selector and final budget. The degree control preserves realized per-donor counts after the same donor/receiver partition while changing spatial correspondence. Seeds and the final-epoch endpoint must be fixed before launch. Existing single-seed results may be reused only after recipe identity is verified. CIFAR standard deviations cannot substitute for ImageNet uncertainty.

Result and interpretation:

Routing / probe	LR (10^{-4})	Best (%)	Final (%)
Global	5.0	77.218	77.216
Flat window 8	5.0	78.290	78.278
Spatial $R = 2$	5.0	78.942	78.942
Spatial $R = 3$	5.0	78.888	78.888
Spatial $R = 3$	7.5	79.316	79.316

Table 1: 224-pixel, 150-epoch routing controls. The first four rows use the same learning rate and base architecture. The last row is the selected higher-rate configuration later used for 300-epoch training. All accuracies are single-seed EMA top-1.

Support	Seeds	Final top-1 mean	Sample SD	Paired Δ to R3
Global	_____	_____	_____	_____
Flat window 8	_____	_____	_____	_____
Degree-matched	_____	_____	_____	_____
Radius 3	_____	_____	_____	_____

Table 2: Reserved ImageNet replication [P04]. Empty cells are missing measurements, not zeros or failed runs. Pairing is by training seed.

5.3 Three-seed CIFAR confirmation

Company ImageNet weights are unavailable, so a local CIFAR-100 campaign tests whether the routing gain repeats across seeds and whether candidate count alone can explain it. The matrix is two selectors $\times$ four geometries $\times$ three seeds, 24 independent 200-epoch runs at 224 pixels with $\lambda = 2$. The primary endpoint, fixed before launch, is epoch-199 EMA top-1 on the 10,000-image test set. Training and evaluation batches are 200. The images are native 32×32 CIFAR frames resized under the historical augmentation protocol; this is not a native high-resolution test.

The four geometries are global routing, flattened window 8, radius 3, and a degree-matched control. The last starts from the realized radius-3 donor/receiver mask after each partition and permutes receiver columns with fixed receiver-ID priorities. Per-donor candidate counts and the receiver degree multiset are preserved; the control is not an independent random graph. The two selectors are the historical batch-range threshold operator and a per-sample float32 bisection operator. Even seed 42 is a new training run (dataloader workers differ from the August sweep).

Table 3 and Figure 3 show that radius 3 is the most accurate geometry under both selectors. All eighteen same-seed radius-3 minus control differences underlying the six rows of Table 4 are positive. Under the historical selector, the mean radius-3 advantages are $+2.11 pp$ versus global, $+1.30 pp$ versus flattened windows, and $+1.42 pp$ versus degree-matched support. The per-sample selector preserves the same ordering, with smaller mean gaps of $+1.63$, $+0.77$, and $+1.19 pp$. The degree-matched gaps indicate that original-grid locality contributes beyond candidate cardinality in this setting. The lower panels of Figure 3 show a persistent mid-to-late training gap rather than a single-epoch spike. Three seeds are enough to show a repeated sign pattern; they are not enough to treat the sample standard deviations as population estimates, and they do not transfer ImageNet uncertainty.

Selector	Global	Flat 8	Degree-matched	$R = 3$
Historical selector	68.48 $\pm$ 0.10	69.29 $\pm$ 0.54	69.17 $\pm$ 0.62	70.59 $\pm$ 0.88
Rowwise selector	68.77 $\pm$ 0.32	69.64 $\pm$ 0.80	69.21 $\pm$ 0.75	70.41 $\pm$ 0.67

Table 3: CIFAR-100 epoch-199 EMA top-1 (%), mean $\pm$ sample standard deviation over seeds 42, 43 and 44. Inputs are resized 32×32 images.

Contrast	Mean Δ (pp)	Sample std	Seeds 42/43/44
Historical $R = 3$ – global	+2.11	0.87	+3.02/+1.28/+2.03
Historical $R = 3$ – flat	+1.30	0.47	+1.83/+1.14/+0.92
Historical $R = 3$ – degree	+1.42	0.30	+1.67/+1.09/+1.51
Rowwise $R = 3$ – global	+1.63	0.51	+1.87/+1.05/+1.98
Rowwise $R = 3$ – flat	+0.77	0.29	+0.50/+0.74/+1.07
Rowwise $R = 3$ – degree	+1.19	0.13	+1.05/+1.22/+1.31

Table 4: Paired same-seed CIFAR-100 differences in percentage points. Every radius-3 comparison is positive. Sample standard deviations use $n - 1$.

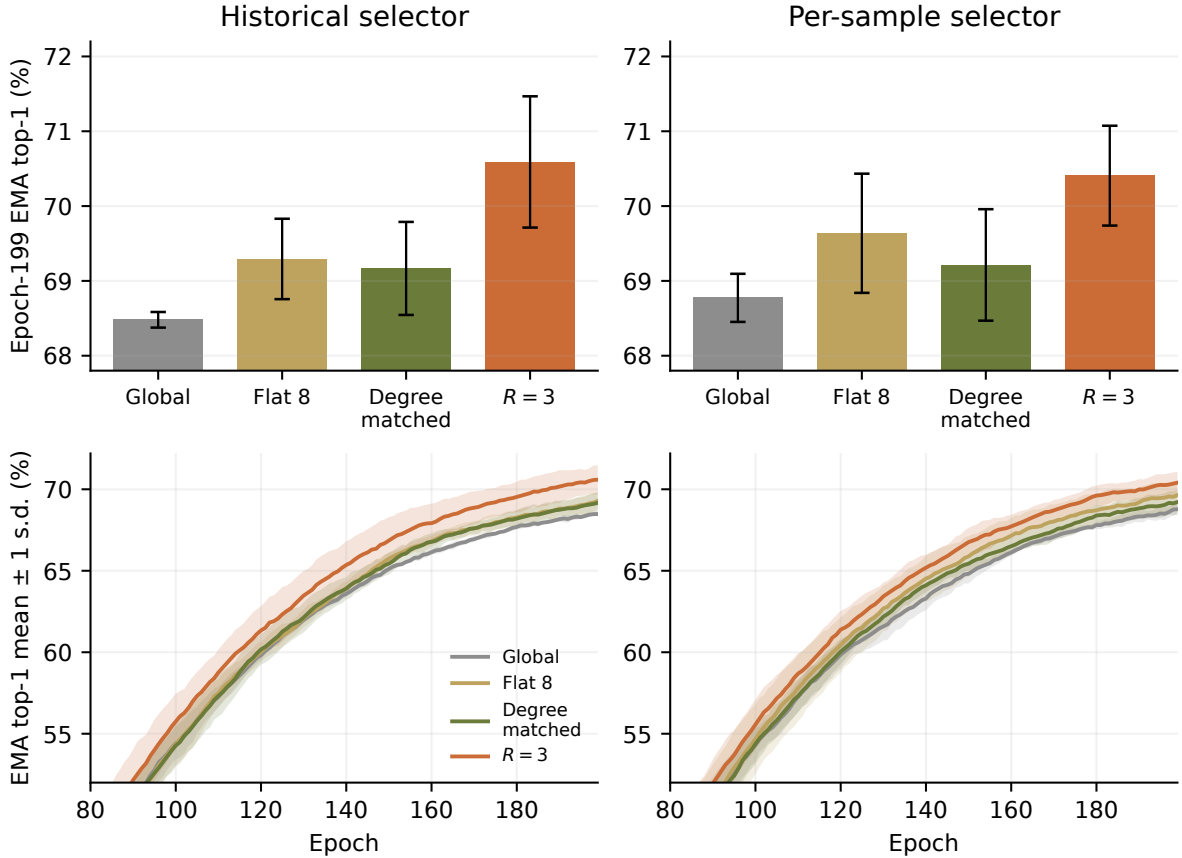


Figure 3: CIFAR-100 three-seed geometry confirmation. Top: epoch-199 EMA top-1 with sample standard-deviation bars. Bottom: mean EMA trajectories with shaded ± 1 sample standard deviation over the same three seeds. Radius 3 exceeds global, flattened, and degree-matched controls under both selectors. These are independent training runs, not evaluation rebatching bars.

5.4 Complete schedules and resolution adaptation

Table 5 reports complete 300-epoch runs and the 384- and 512-pixel fine-tunes. At 224 pixels, MergeNet trails dense DeiT-S/8 by 0.878 *pp* using best checkpoints and 0.812 *pp* using final checkpoints. It exceeds DeiT-S/16 by 1.676 *pp*, but the latter is a much cheaper model at this input resolution. The higher accuracy relative to coarse patches does not establish better efficiency.

After 30 epochs at 384 pixels, radius-3 MergeNet reaches 82.582%, compared with 82.976% for dense DeiT-S/8. Its best-checkpoint gain over its own 224-pixel result is 1.214 *pp*, while dense DeiT gains 0.730 *pp*. The gap therefore narrows to 0.394 *pp*; using final checkpoints it is 0.324 *pp*. Scaling the radius to 5.143 produces 82.574%, a difference of -0.008 *pp* from radius 3. This one comparison gives no evidence of a useful accuracy gain from that particular scaling rule, but cannot rule out other radii or recipes.

A further 15 epochs at 512 pixels reach 83.068% for dense DeiT and 82.618% for MergeNet. The gap is 0.450 *pp*, slightly wider than at 384 pixels. Table 6 and Figure 4 summarize the series. Dense DeiT gains 0.092 *pp* from 384 to 512 pixels. MergeNet gains 0.044 *pp* over its actual radius-5.143 parent (82.574%); the difference from the separately selected radius-3 384-pixel result is 0.036 *pp*. The latter is a comparison of selected configurations, not a within-branch fine-tuning gain. The 384-pixel narrowing is therefore not the start of a monotone closing trend.

Figure 5 shows that the models reach their best 384-pixel scores at different epochs: 7 for dense DeiT, 20 for radius 3, and 22 for radius 5.143. Both 512-pixel runs peak at epoch 6 of 15. These observations do not prove that the router requires three times as much adaptation, nor that truncating a schedule would reproduce a newly scheduled shorter run. Short terminal slopes cannot justify extrapolating another 300 training epochs.

5.5 Optimizer probes do not recover the dense gap

The selected 150-epoch radius-3 configuration reaches 79.316%. Before the later probes ran, a promotion rule required a gain larger than 0.40 *pp* (above 79.716%) to promote a configuration. This is a campaign decision threshold, not an estimated noise level, statistical null hypothesis, or equivalence margin. Table 7 and Figure 6 report the completed one-at-a-time tests.

Longer warmup changes the early-training stall—the epoch-24 to 36 gain rises from 6.99 *pp* to 22.49 *pp*—but finishes 0.078 *pp* below the selected arm. A $\lambda = 1.5 \rightarrow 2$ curriculum finishes 0.006 *pp* below it. Raising the learning rate to 10^{-3} gains 0.034 *pp*. Reducing drop-path from 0.1 to 0.07, which matches the mean per-block stochastic depth of a single twelve-block DeiT ramp, is the largest completed movement (+0.218 *pp*) and still misses the promotion band. The incomplete weight-decay snapshot sits 0.262 *pp* below the selected arm after 140 epochs. The remaining ten epochs have no established accuracy upper bound, so the run cannot be classified as a completed negative result. The completed probes provide no evidence that these particular changes recover the gap; they do not rule out optimization effects in other recipes or at 300 epochs.

A test-time re-evaluation of the 300-epoch MergeNet checkpoint gives a related boundary (Table 8). Using `method=none` records 81.354% at an achieved 392-token sequence, 0.006 *pp* below the native MergeNet evaluation of 81.360%. Raising the evaluation budget to 523 tokens yields 81.380%; lowering it to 196 tokens yields 80.228%. These are correlated evaluations of one checkpoint. They do not show that training-time compression is free. They do show that modest evaluation-budget changes do not recover the dense-DeiT gap. The `method=none` cell still achieves 392 tokens, so it is not a 784-token dense readout of the local encoder.

5.6 Representation initialization: reserved endpoint

Evidence pending [P01]: Complete the interrupted initialization experiment

The 105-epoch record is partial; its observed maximum is not the 150-epoch endpoint.

The interrupted arm initializes from the 300-epoch DeiT-S/8 `model_best.pth.tar` file. Its startup log reports 152 loaded tensors, 25 missing tensors and no unexpected tensors. This is a partial mapping, not evidence that every local and latent parameter inherits the dense model. The checkpoint was selected using EMA accuracy, but the available loader prefers `state_dict`. The actual state key and tensor fingerprint must be confirmed before calling this raw or EMA initialization. The local encoder remains trainable; this is not router-only training.

Model	Pixels	Epochs	Best (%)	Final (%)	Best epoch
DeiT-S/16	224	300	79.692	79.684	290
DeiT-S/8	224	300	82.246	82.162	277
MergeNet, $R = 3$	224	300	81.368	81.350	291
DeiT-S/8	384	300+30	82.976	82.848	7
MergeNet, $R = 3$	384	300+30	82.582	82.524	20
MergeNet, $R = 5.143$	384	300+30	82.574	82.532	22
DeiT-S/8	512	300+30+15	83.068	83.038	6
MergeNet, $R = 6.857$	512	300+30+15	82.618	82.480	6

Table 5: Complete main runs. Fine-tunes initialize from the corresponding EMA checkpoint: 384-pixel runs from 300-epoch 224-pixel weights, and 512-pixel runs from the 384-pixel stage. The 512-pixel MergeNet run uses the radius-5.143 384-pixel EMA file and $R = 6.857$. The selected MergeNet scratch configuration uses a higher learning rate than dense DeiT; the fine-tuning recipe is shared. Best and final checkpoints are kept distinct; fine-tuning best-epoch indices are relative to that stage.

Pixels	Schedule	DeiT-S/8 (%)	MergeNet (%)	Gap (pp)
224	300 scratch	82.246	81.368	-0.878
384	300+30	82.976	82.582	-0.394
512	300+30+15	83.068	82.618	-0.450

Table 6: Dense DeiT-S/8 versus the selected MergeNet configuration across the completed resolution series. MergeNet at 512 pixels is the radius-scaled branch described in Table 5.

The existing scratch reference with matching drop-path 0.07 is `s5_mn_dp07_150e`, whose best EMA top-1 is 79.534%, subject to a check of all other resolved arguments. The drop-path-0.1 selected arm is not an initialization-only control. Report the source model’s 300-epoch pretraining separately from 150-epoch adaptation. Resume the interrupted arm’s own last checkpoint, retaining optimizer, scheduler, EMA and scaler state; a fresh initialization would be a new run.

Result and interpretation:

5.7 Progressive latent compression: reserved comparison

Evidence pending [P02]: Test the queued schedule at a matched final budget

The configuration exists; no completed run or measured layerwise trace is present.

The candidate retains the full-grid local encoder and the 784-to-392 gather. It requests 192 additional removals across six latent blocks, with nominal patch counts $392 \rightarrow 360 \rightarrow 328 \rightarrow 296 \rightarrow 264 \rightarrow 232 \rightarrow 200$. The class token is additional. A runtime trace must confirm attention input, merge output and MLP input counts. This candidate does not eliminate the initial gather or shorten the earlier local blocks.

Comparing this arm with the native 392-patch model mixes schedule and final budget. For schedule attribution, add a single-gather control at the same achieved 200-patch endpoint, with matched depth, initialization, training recipe and drop-path 0.07. Match ToMe’s achieved budget for deployment comparisons while recording its different training history.

Result and interpretation:

5.8 Compression after training

Figure 7 compares full-validation accuracy at equal final patch counts. At 224 pixels and 392 retained patches, ToMe achieves 81.932%, compared with MergeNet’s re-evaluated 81.360%. At 196 patches, the scores are 81.752% and 80.228%, a 1.524 pp difference. The latter MergeNet point changes the budget of the $\lambda = 2$ checkpoint at evaluation; it is not the separately trained, structurally different $\lambda = 4$ model.

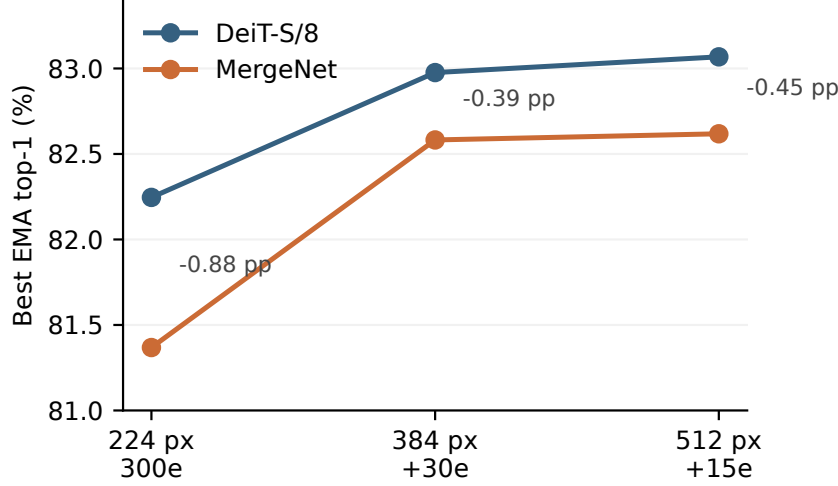


Figure 4: Best EMA top-1 at 224, 384 and 512 pixels. Annotations are MergeNet minus dense DeiT-S/8. The 384-pixel stage is the closest of the three, not a point on a continuing improvement trend.

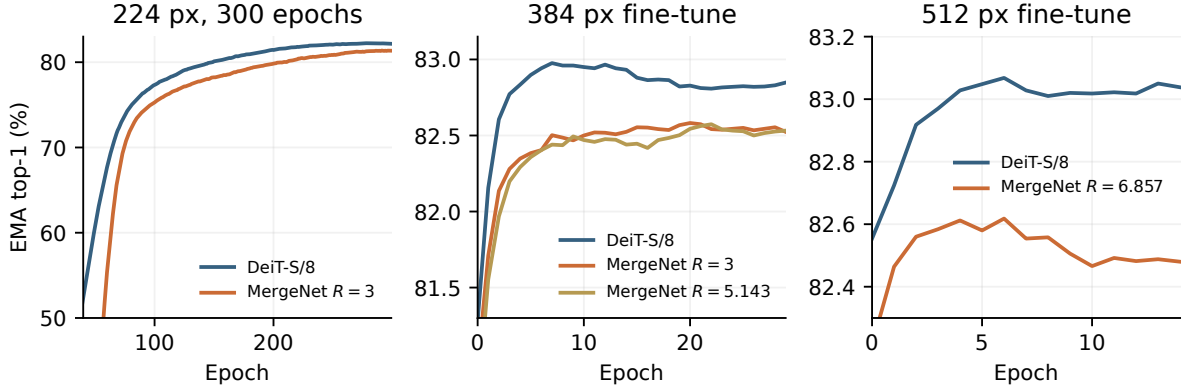


Figure 5: Recorded EMA validation curves. Left: epochs 40–299 of complete 300-epoch runs. Middle: all 30 fine-tuning epochs at 384 pixels. Right: all 15 fine-tuning epochs at 512 pixels. Curves share a seed and are not confidence intervals. Later peak epochs do not identify why adaptation occurs or predict a longer schedule.

For 384-pixel fine-tuned checkpoints, ToMe reaches 82.866% at 1,152 patches, versus 82.562% for MergeNet, and 82.742% at 576 patches, versus 81.300%. Across all twenty matched budget points—five at native 224 pixels, five each at 288 and 384 pixels without resolution fine-tuning, and five after 384-pixel fine-tuning—ToMe has higher recorded accuracy. The differences range from 0.304 *pp* to 1.524 *pp*. These points are correlated evaluations of a small number of checkpoints, not twenty independent training trials. PiToMe is also competitive, but does not exceed MergeNet at every point; for example, at 1,152 patches after fine-tuning its 82.534% is slightly below MergeNet’s 82.562%.

Sweep values are re-evaluations and need not equal the training summary exactly. For example, the native 224-pixel MergeNet sweep records 81.360% versus a training best of 81.368%. We preserve both measurements rather than rounding them into agreement. Repeated PiToMe evaluations also differ by up to 0.034 *pp* in the supplied record. This is variation in repeated evaluation, not a measured across-seed training variance or a universal noise threshold. Most importantly, these accuracy results alone do not establish a joint accuracy–latency advantage; Section 6 explains the harness mismatch that prevents that conclusion.

5.9 Training-budget isolation: reserved ablation

Evidence pending [P07]: Separate compression strength from architecture changes

This extension is required for a training-compression causal claim, not for the existing checkpoint budget sweep.

150-epoch probe	LR (10^{-4})	Best (%)	Δ vs selected	Status
Spatial $R = 3$ (selected)	7.5	79.316	+0.000	Complete
Warmup 20	7.5	79.238	-0.078	Complete
λ 1.5 $\rightarrow$ 2 curriculum	7.5	79.310	-0.006	Complete
Learning rate 10^{-3}	10	79.350	+0.034	Complete
Drop-path 0.07	7.5	79.534	+0.218	Complete
Weight decay 0.03	7.5	79.054	-0.262	140/150

Table 7: 150-epoch probes off the selected radius-3, learning-rate 7.5×10^{-4} configuration. Δ is best EMA top-1 minus 79.316. The weight-decay run was stopped at epoch 140 by a scheduled reboot; it is not a completed endpoint.

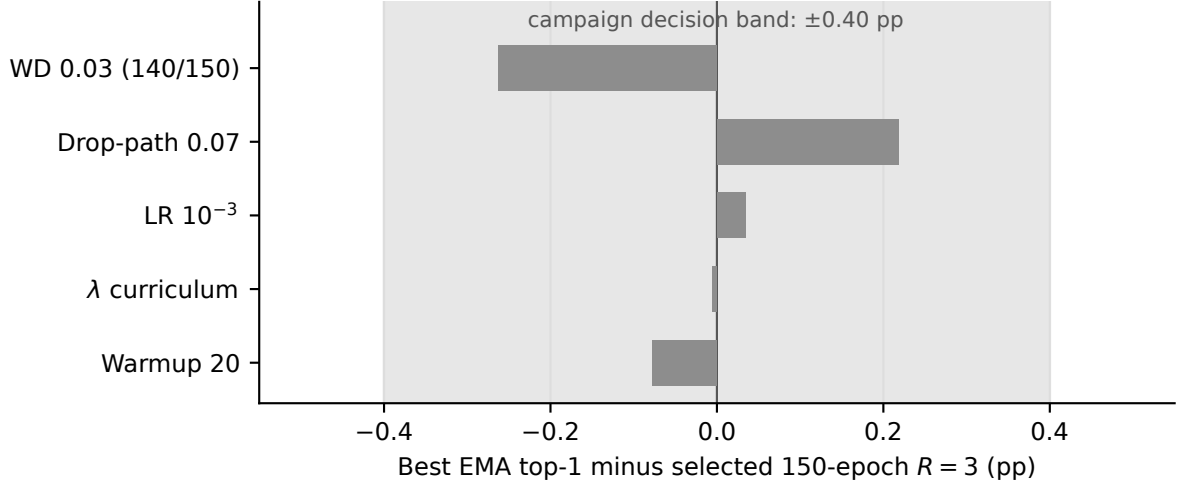


Figure 6: Completed 150-epoch optimizer probes relative to the selected radius-3 run. The shaded interval is the campaign-defined ± 0.40 pp decision band. The weight-decay bar uses a 140-epoch snapshot.

Train each budget with identical local/latent depths, local attention window, width, recovery path, selector and optimization recipe. Verify integer carrier counts on the instantiated model. A no-reduction control must execute a verified no-compression path: the existing `method=none` cell still has 392 patches. Record any code changes required for that control. Select budgets before inspecting validation outcomes and use paired seeds.

Result and interpretation:

Evaluation of 300-epoch MN checkpoint	Tokens	Top-1 (%)	vs native
Native method	392	81.360	0.000
method=none	392	81.354	-0.006
Native method, 523 carriers	523	81.380	+0.020
Native method, 196 carriers	196	80.228	-1.132

Table 8: Full-validation re-evaluations of the 300-epoch radius-3 checkpoint at 224 pixels. Native method at 392 tokens is 81.360% here versus a training-summary best of 81.368%; both measurements are retained.

Arm	Best EMA	Final EMA	Best epoch	State receipt
DeiT-initialized, completed 150 epochs	_____	_____	_____	_____
Difference to verified scratch	_____	_____	_____	_____

Table 9: Reserved initialization results [P01]. Accuracy is in percent and differences in percentage points. Partial-curve values are not endpoints.

Arm	Final patches	Best EMA	Final EMA	Batch latency (ms)
Progressive latent	_____	_____	_____	_____
Single gather, same endpoint	_____	_____	_____	_____
ToMe, same endpoint	_____	_____	_____	_____

Table 10: Reserved schedule comparison [P02]. Requested budgets do not substitute for achieved counts. Timing requires the common protocol in P03.

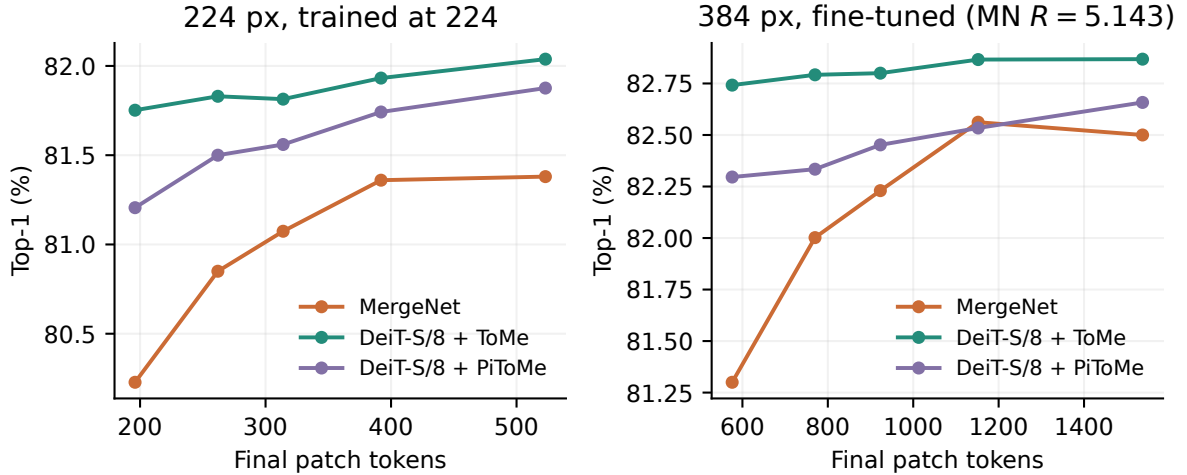


Figure 7: Accuracy-only budget sweeps. Left: 300-epoch 224-pixel checkpoints. Right: 384-pixel fine-tuned checkpoints, with radius 5.143 for MergeNet. ToMe and PiToMe require no additional training on top of the trained dense backbone. Lines connect measured points, not confidence bands. Equal final patch counts do not match the full computation trajectory.

Training condition	Patches	Seeds	Final EMA mean / SD	GMAC / ms
Verified no physical reduction	_____	_____	_____	_____
Native compression	_____	_____	_____	_____
Stronger compression	_____	_____	_____	_____

Table 11: Reserved controlled compression ablation [P07]. The historical compound $\lambda = 4$ run cannot fill an isolated budget row.

6 Execution and Implementation Diagnostics

6.1 Arithmetic savings do not guarantee lower latency

The 224-pixel analytic estimate is 14.13 GMAC per image for MergeNet and 22.41 GMAC for dense DeiT-S/8, a 36.9% reduction. This estimate counts multiply-accumulates as one unit, includes projections, MLPs, attention matmuls, routing transport, and recovery, and does not assign MACs to sorting, indexing, copying, or threshold selection. It is an arithmetic accounting model rather than a measured instruction count. Figure 8 shows why a final carrier budget cannot describe all computation.

The guarded synthetic benchmark nevertheless measures MergeNet inference at 45.71 ms per batch, versus 42.68 ms for dense DeiT-S/8 (Table 12). MergeNet takes 7.1% more inference time and 3.5% more training-step time in this experiment. It also uses more peak allocated memory. DeiT-S/16 is substantially faster, reinforcing that the accuracy gain from retaining a finer input grid has a real execution cost.

Several implementation properties offer plausible explanations. Full-grid local MLPs and projections remain, sparse routing introduces indexed access and reductions, the threshold operator performs sorting and cumulative operations, and global recovery adds another attention operation. However, the delivered decomposition uses CPU operator profiles and analytic byte estimates. Its dispatch counts are not measured CUDA kernel-launch counts, and its logical tensor traffic is not measured HBM traffic. We therefore do not infer a proven memory-bandwidth bottleneck, a measured arithmetic intensity, or an end-to-end speedup from a proposed kernel optimization.

At 384 pixels, the guarded batch-64 radius-3 benchmark measures 136.54 ms for MergeNet inference and 170.61 ms for dense DeiT, a 20.0% reduction in batch time. A batch-128 run measures 262.02 versus 324.36 ms, a 19.2% reduction; its training-step times are 889.76 versus 1135.48 ms. Thus the measured cost balance changes with resolution. These architecture benchmarks use random initialization and remain distinct from checkpoint accuracy. A later radius-5.143 latency sweep measures ToMe with proportional attention *disabled*, unlike the accuracy sweep. It also uses slightly different achieved patch counts and retains token-counting hooks during timing. Appendix B reports these cells with their actual settings. The available record does not support attaching the earlier accuracy scores to these times or declaring that one method dominates the other on both axes. A common harness must use the same attention policy, actual budget, checkpoint identity, precision and hook-free timed path before making that claim.

6.2 Soft selection depends on batch composition

A separate CPU probe identifies a functional issue in the inherited threshold-based soft top- k implementation. For an input tensor $x \in \mathbb{R}^{B \times N}$, it normalizes using

$$\hat{x}_{bi} = 30 x_{bi} / \left(\max_{b',j} x_{b'j} - \min_{b',j} x_{b'j} \right). \quad (10)$$

The range is computed across the entire batch. Consequently, selecting weights for one sample can depend on unrelated samples. Holding the first row fixed at (1, 2, 3, 4) and $k = 2$, changing only the second row from (1.2, 2.2, 3.2, 3.8) to (10, 20, 30, 40) changes a first-row soft weight by up to 0.336987. A loss depending only on the first row yields a nonzero gradient on the second row, with maximum magnitude 0.017355 in the probe. Both float32 and float64 exhibit the effect. An entirely constant batch also makes the unclamped denominator zero, producing non-finite outputs and gradients.

This operator is used directly in the routing assignment, so the issue is not limited to the final selection’s straight-through gradient. In particular, changing evaluation batch composition can change soft routing even in `eval` mode. The probe establishes the operator-level dependency; it does not establish how much any supplied ImageNet checkpoint’s top-1 would change, or explain the small sweep-versus-summary discrepancies by itself. The main same-recipe geometry controls remain comparisons under their recorded batch conditions. A corrected implementation requires its own checkpoint evaluation and training evidence rather than inheriting the reported scores without verification.

Appendix G adds independent tests that do not require the company ImageNet weights. A common A100 harness controls ToMe’s attention policy and verifies exact patch budgets; it does not recover the H20 crossover on the local stack. Existing local CIFAR weights also show that batch composition changes predictions, while the tested radius-3 advantage persists. These follow-ups narrow the interpretation without replacing the historical ImageNet measurements.

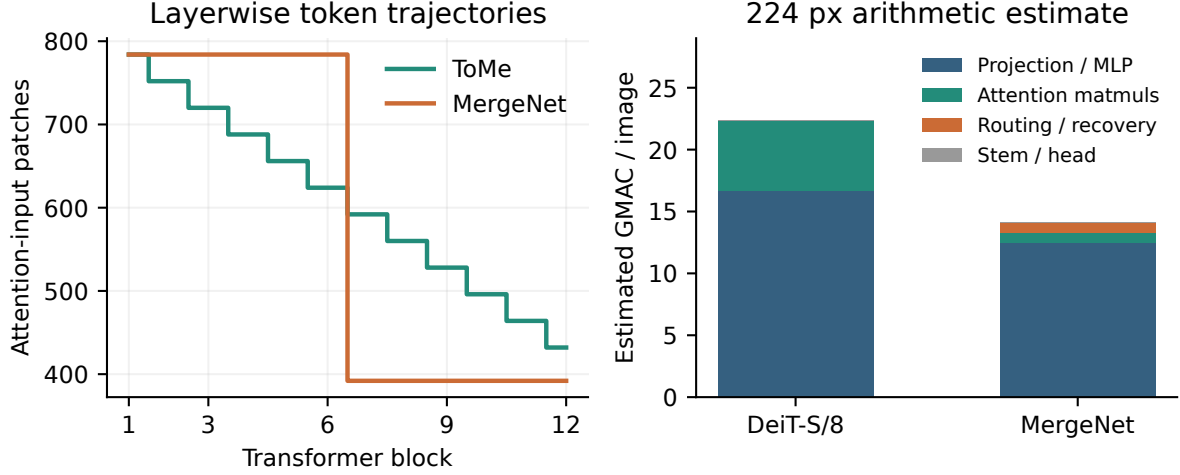


Figure 8: Two code-derived views of cost. Left: the attention-input patch trajectory at a final budget of 392; ToMe merges within blocks, while MergeNet physically reduces the sequence between stages. ToMe’s MLP inputs can be shorter than its attention inputs. Right: analytic MAC components, with non-arithmetic operations omitted from the count. Recovery is included in the MergeNet routing/recovery component. These are not measured GPU profiles.

Model	Params (M)	Train (ms)	Infer (ms)	IQR (ms)	Train GiB	Infer GiB
DeiT-S/16	22.05	37.62	9.99	0.005	2.472	0.367
DeiT-S/8	22.06	149.56	42.68	0.099	8.555	0.799
MergeNet, $R = 3$	23.10	154.87	45.71	0.073	10.071	1.125

Table 12: September 5, 2026, 224-pixel batch-64 synthetic H20-3e benchmark, with random model initialization and fp16 autocast. Train time includes forward, loss, backward and AdamW update. Infer time is a forward-only median; the IQR column refers to inference. Memory is peak *allocated* GiB. These model-execution cells are separate from the trained-checkpoint accuracy experiments.

6.3 Trained-checkpoint joint evaluation: reserved frontier

Evidence pending [P03]: Pair accuracy and timing from one verified configuration

The weights can remain on the collaborator’s machine; return only permitted evaluation records and figures.

For each dense, MergeNet, ToMe and PiToMe cell, bind a checkpoint fingerprint and state key to the same resolved arguments, resolution, preprocessing, precision, batch composition and achieved patch budget. ToMe proportional attention must agree between accuracy and timing. Count class tokens separately, verify per-layer shapes, and remove hooks before timing. Evaluate all 50,000 validation images. For the batch-dependent historical selector, fix evaluation batches, ordering and distributed partitioning.

Measure on an otherwise idle device. Archive warmup/repetition counts, synchronized CUDA-event samples, medians and IQRs. Report allocated and reserved memory separately, and specify whether latency includes data loading. Synthetic-input measurements remain labeled as model-execution timings. Do not join the old accuracy sweep to another random-weight configuration.

Method	Patches	EMA top-1	Median ms	IQR ms	Allocated GiB
Dense DeiT-S/8	_____	_____	_____	_____	_____
MergeNet	_____	_____	_____	_____	_____
ToMe	_____	_____	_____	_____	_____
PiToMe	_____	_____	_____	_____	_____

Table 13: Reserved trained-checkpoint joint cells [P03]. Repeat this block for each resolution/budget and attach device, batch, state key and attention policy. All numerical fields remain unfilled.

P03: measured accuracy versus batch latency at 224 and 384 pixels
Reserved space; no measurements or images supplied.

Figure 9: Reserved joint comparison [P03]. No dominance ordering is assumed.

Result and interpretation:

6.4 ImageNet selector transfer: reserved experiment

Evidence pending [P05]: Separate batch invariance from learning behavior

Completed CIFAR runs do not determine ImageNet accuracy after a selector change.

Hold selected images fixed while replacing only their batch companions. Compare exact repeats, original selection and rowwise selection; export logit differences, changed classes, selection-mass errors and finite-gradient checks. A learning claim additionally requires matched ImageNet training. Changing the operator in old weights is an inference intervention. The rowwise implementation changes normalization and threshold solution together; attributing an effect to one component needs additional factorial controls. Generalizing the issue to other implementations requires their own audits.

Condition	Top-1	Max logit change	Changed classes	Finite checks
Original selector, old weights	_____	_____	_____	_____
Rowwise intervention, same weights	_____	_____	_____	_____
Matched rowwise training	_____	_____	_____	_____

Table 14: Reserved ImageNet selector tests [P05]. Image population, batches, seeds and state identities accompany each distinct experimental condition.

Result and interpretation:

6.5 Execution-cost attribution: reserved profile

Evidence pending [P08]: Measure device activity and complete training steps

Needed for a specific bottleneck explanation or training-speed claim.

Profile local attention/MLPs, selection, transport, gather, recovery and latent blocks on one fixed software stack. Device operator times may overlap and must not be assumed additive. HBM bandwidth claims require device counters, not logical tensor sizes. Compare layouts with identical inputs, loss, backward, optimizer and AMP policy; report actual updates and skipped steps. Long-run accuracy or stability needs separate matched training. The existing approximate 5% inference gain does not fill these training cells.

Result and interpretation:

Implementation	Step median ms	IQR ms	Allocated GiB	Actual updates
Original layout	_____	_____	_____	_____
Layout variant	_____	_____	_____	_____

Table 15: Reserved complete-step timing [P08]. Per-region device profiles and any long-run checks are separate evidence.

7 Discussion and Limitations

The results distinguish an internal architectural improvement from a competitive compression system. Original-grid support improves this architecture relative to its own unconstrained router, and a three-seed CIFAR campaign finds the same sign against global, flattened, and degree-matched controls. Yet the full ImageNet model loses accuracy to the selected dense baseline at 224, 384 and 512 pixels, its 224-pixel implementation is slower in guarded synthetic tests, and post-training ToMe provides a stronger accuracy baseline at every matched budget in the sweep. Optimizer-side 150-epoch probes do not close the selected-configuration gap. These findings motivate revisiting the computation path, the compression schedule, and representation initialization; they do not show that learning a merge criterion is intrinsically inferior.

The evidence is limited in several ways. All ImageNet training runs use seed 42, and the validation set serves both checkpoint selection and reporting. We have no across-seed ImageNet uncertainty estimate or ImageNet per-image predictions for paired significance tests. Appendix G provides predictions for two independently replayed local CIFAR checkpoints from an earlier campaign; Section 5.3 adds three-seed training on resized CIFAR-100, which is not native-resolution ImageNet. Geometry and candidate cardinality remain coupled on ImageNet. The 300-epoch MergeNet configuration benefits from an explicit learning-rate screen, while the dense baseline is not subjected to an equally broad screen. The $\lambda = 4$ training variant changes multiple architectural parameters, and test-time budget changes do not replace a clean training-budget ablation. The 512-pixel MergeNet run is the radius-scaled branch seeded from the radius-5.143 384-pixel EMA file, not a radius-3 continuation.

The experiment artifacts also bound reproducibility. Raw metrics, resolved arguments, launch records, environment information, benchmark scripts and checkpoint receipts are available in the campaign packet. Company policy prevents transfer of the trained ImageNet weights, and a release commit alone does not capture every change in the collaborator’s working tree. The companion analysis preserves the source identities and distinguishes recorded results from independent operator diagnostics. The training environment uses PyTorch 2.7.1 with CUDA 11.8, timm 0.9.11 and FlashAttention 2.7.3, which differs from the local release’s pinned environment. CPU profiles and H20 synthetic timings cannot establish performance on every accelerator or under deployment load. The local follow-up resolves the ToMe attention-policy and patch-count mismatch for synthetic architecture timing. A verified trained-checkpoint joint frontier remains unavailable.

Two ImageNet runs in the September 19 packet are unfinished. The weight-decay probe stopped at epoch 140. A DeiT-initialized MergeNet run reached epoch 104 of 150 and should be resumed from its last checkpoint; we do not treat its partial curve as an endpoint, because earlier mid-training leads in this campaign have decayed. Progressive latent merging was queued and not started. A DeiT-B/16 300-epoch run overfits under drop-path 0.1 (Appendix D) and is not a usable base-scale anchor.

The most informative remaining experiments follow directly from these limits. Resume the interrupted DeiT-initialized run to a 150-epoch endpoint, and run the queued progressive-latent configuration as its own family. Any new local ImageNet training must be reported as a separate experiment from the company packet. Repeat the global, flattened, radius-3 and degree-matched controls at additional ImageNet seeds only after the representation-level tests are complete. The layout follow-up supplies bounded parity and complete-model inference timing; complete training-step timing and long-run behavior remain open. These experiments would test concrete explanations for the observed gaps; merely extending a schedule based on a short terminal slope would not.

8 Conclusion

Spatially restricting soft token routing improves the recorded accuracy of MergeNet relative to its global and flattened-window controls. The ImageNet benefit survives both best- and final-checkpoint reporting. A three-seed CIFAR-100 campaign reproduces a radius-3 advantage under two selectors and against a

degree-matched support control. Higher-resolution fine-tuning narrows the ImageNet gap to dense DeiT-S/8 at 384 pixels, but a completed 512-pixel stage does not continue that closing. Completed optimizer probes do not exceed the campaign’s improvement threshold over the selected 150-epoch configuration. The composed system still does not establish a superior compression method. Full-grid processing before the gather, recovery cost, and routing overhead separate nominal compression from execution savings. The measured dense-relative latency balance improves at higher resolution on H20, while the independent A100 stack shows no dense-relative speedup at either tested resolution. A unified local timing protocol and CIFAR checkpoint replay narrow the evidence: the layout change gives modest whole-model gains, and spatial-routing accuracy benefits persist despite observable batch dependence. Strong post-training accuracy baselines and unavailable ImageNet weights still limit deployment claims. The resulting study provides a reproducible account of the measured tradeoffs, a three-seed routing confirmation, and a specific batch-dependence diagnostic, while leaving a corrected, checkpoint-verified accuracy–latency frontier and the interrupted representation-initialization run as explicit open results.

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A Training Recipe and Evidence Inventory

Table 16 summarizes the resolved run arguments. The base width is 384 with six heads and MLP expansion four. Stochastic depth is configured at 0.1; MergeNet constructs separate local and latent ramps, unlike a single twelve-block dense ramp. The global and flattened routing controls use their dense/historical scoring paths, while radius-2/3 routing uses the sparse spatial backend; the routing support and its numerical implementation should therefore both be recorded. EMA validation replaces raw-model metrics before the summary row is written, so `eval_top1` is an EMA metric.

Setting	224-pixel scratch	384/512-pixel fine-tune
Epochs	150 or 300	30 at 384; 15 at 512
Effective global batch	1024	1024
Optimizer	AdamW	AdamW
Learning rate	5×10^{-4} or 7.5×10^{-4}	10^{-5}
Minimum learning rate	10^{-5}	10^{-6}
Weight decay	0.05	10^{-8}
Warmup epochs	5 (probe: 20)	1
Warmup learning rate	10^{-6}	10^{-6}
EMA decay	0.99996	0.9996
Mixup / CutMix	0.8 / 1.0	0 / 0
Label smoothing	0.1	0.1
Random erasing	0.25	0.25
RandAugment	<code>rand-m9-mstd0.5-inc1</code>	same
Gradient clip norm	1.0	1.0
Validation crop percentage	0.9	1.0
MergeNet auxiliary-logit weight	ramp to 0.05	0.05
Teacher / feature / routing distillation	disabled	disabled

Table 16: Main recipe. Launch records take precedence over configuration-file defaults, especially for microbatch size and gradient accumulation.

The 150-epoch MergeNet runs start the auxiliary-logit contribution at epoch 10 and ramp it over 10 epochs; the 300-epoch run uses epoch 20 and a 20-epoch ramp. Fine-tunes enable weight 0.05 from the start. The recorded environment is Python 3.10.20, PyTorch 2.7.1+cu118, torchvision 0.22.1+cu118, timm 0.9.11, FlashAttention 2.7.3 and Triton 3.3.1 on H20-3e GPUs. Both four and eight ranks divide 50,000, avoiding validation-sampler padding for this split. Other world sizes require a separate check.

Three nominal launcher failures occurred after complete metric histories: the 150-epoch DeiT-S/16, the compound $\lambda = 4$ run, and dense 512-pixel fine-tuning. Complete summaries and supplied checkpoint receipts distinguish them from early failures. Receipts record checks on another machine, not weights independently loaded in this audit. The λ curriculum and 512-pixel MergeNet fine-tune are now complete 150-epoch and 15-epoch records. The weight-decay probe and DeiT-initialized run remain partial.

Run ID	Epochs	Status	Best top-1
s2_deit_p16_150e	150/150	Complete	77.196
s2_deit_p8_150e	150/150	Complete	79.992
s2_mn_flat_w8_150e	150/150	Complete	78.290
s2_mn_global_150e	150/150	Complete	77.218
s2_mn_l4_150e	150/150	Complete	78.366
s2_mn_r2_150e	150/150	Complete	78.942
s2_mn_r3_150e	150/150	Complete	78.888
s2_mn_r3_lr75_150e	150/150	Complete	79.316
s3_deit_p16_300e	300/300	Complete	79.692
s3_deit_p8_300e	300/300	Complete	82.246
s3_mn_r3_lr75_300e	300/300	Complete	81.368
s3_deit_p8_ft384_30e	30/30	Complete	82.976
s3_mn_r3_ft384_30e	30/30	Complete	82.582
s3_mn_r5_ft384_30e	30/30	Complete	82.574
s4_mn_lamcurr_150e	150/150	Complete	79.310
s4_mn_warm20_150e	150/150	Complete	79.238
s4_deit_ft512_15e	15/15	Complete	83.068
s4_mn_ft512_15e	15/15	Complete	82.618
s5_mn_dp07_150e	150/150	Complete	79.534
s5_mn_lr1e3_150e	150/150	Complete	79.350
s5_mn_wd03_150e	140/150	Partial	79.054
s6_deitB_p16_300e	300/300	Complete	80.442
s7_mn_deitinit_150e	105/150	Partial	79.206

Table 17: Status in the September 19 packet. Partial metrics are included for accounting only and are not endpoints. The DeiT-B best score is a pre-overfitting peak (Appendix D). The companion data retain full precision and source summary lines.

B Detailed Budget and Timing Comparisons

Table 18 gives the values behind the main accuracy panels. The MergeNet checkpoint uses radius 3 at 224 pixels and radius 5.143 after 384-pixel fine-tuning. Final patch budgets match within a row, but layerwise attention and MLP inputs do not. ToMe and PiToMe shorten sequences within successive blocks, while MergeNet gathers once between stages. The companion CSV also contains the two resolution-transfer panels using 224-pixel checkpoints without fine-tuning.

Checkpoint / pixels	Patches	MergeNet	ToMe	PiToMe	Δ ToMe
224 trained / 224	523	81.380	82.038	81.876	+0.658
224 trained / 224	392	81.360	81.932	81.742	+0.572
224 trained / 224	314	81.074	81.814	81.560	+0.740
224 trained / 224	262	80.850	81.830	81.500	+0.980
224 trained / 224	196	80.228	81.752	81.206	+1.524
384 tuned / 384	1537	82.500	82.868	82.658	+0.368
384 tuned / 384	1152	82.562	82.866	82.534	+0.304
384 tuned / 384	923	82.230	82.800	82.452	+0.570
384 tuned / 384	770	82.002	82.792	82.334	+0.790
384 tuned / 384	576	81.300	82.742	82.296	+1.442

Table 18: Top-1 percentages at equal final patch counts; Δ is ToMe minus MergeNet in percentage points. These values cannot be paired with Table 19 to construct an accuracy-latency frontier.

At the approximately half-size operating point, ToMe processes 1,151 patches and MergeNet 1,152, taking 129.810 and 155.255 ms respectively. These times do not prove a joint ordering for the accuracy sweep’s ToMe setting. A common harness should use exactly 1,152 patches with proportional attention enabled, verify the checkpoint and preprocessing, and remove shape hooks before timing. An earlier latency benchmark silently clamped several requested ToMe budgets to the same achieved count; those obsolete cells are excluded.

Separate 384-pixel radius-3 benchmarks remain distinct from the radius-5.143 sweep. Speed percentages at different batch sizes, radii or harness settings should not be averaged into an unexplained range. Peak reserved memory and peak allocated memory also measure different quantities.

ToMe patches	MN patches	ToMe ms	MN ms	ToMe GiB	MN GiB
1536	1538	147.30	178.17	2.793	3.619
1151	1152	129.81	155.25	2.764	3.564
922	923	122.44	143.02	2.746	3.721
769	770	118.17	136.59	2.738	3.719
575	576	106.45	129.38	2.723	3.557

Table 19: Separate 384-pixel synthetic latency experiment: batch 64, H20-3e, fp16 autocast, random initialization, MergeNet radius 5.143, and ToMe proportional attention disabled. Patch counts exclude the class token here. Memory is peak *reserved* GiB, unlike Table 12. Token-counting hooks remain active during timing.

C Operator-Level Diagnostic and Reproduction Boundaries

The batch-dependence probe imports only the released threshold-selection function. It does not instantiate an ImageNet model or load a checkpoint. A fixed first row is evaluated alone and with controlled second rows; a weighted loss on the first row tests cross-sample gradients. Constant-input cases test finite forward and backward behavior. These probes establish operator properties, not an accuracy gain from replacing the operator.

Per-sample normalization removes the demonstrated coupling, but does not by itself validate the closed-form threshold solver over every budget and tie pattern. A patched closed-form candidate passes 116 of 120 tensor coverage cases and is not recommended for integration. A separate slow reference finds a per-row threshold t for normalized scores z satisfying

$$\sum_i h(z_i - t) = k, \quad h(\delta) = \begin{cases} 1 - \frac{1}{2} \exp(-\delta), & \delta > 0, \\ \frac{1}{2} \exp(\delta), & \delta \leq 0. \end{cases} \quad (11)$$

For $0 < k < N$, the sum is continuous and strictly decreasing in t . With $v_i = \partial \mathcal{L} / \partial h_i$, implicit differentiation gives

$$\nabla_{z_i} \mathcal{L} = h'_i \left(v_i - \frac{\sum_j v_j h'_j}{\sum_j h'_j} \right). \quad (12)$$

PyTorch then propagates gradients through the per-row normalization to the raw scores. The reference normalizes per row, solves in float64 with 60 bisection steps, and casts the result back to the input dtype. It passes all 120 coverage cases for finite forward/backward values and selection mass, plus three nondegenerate gradient checks. Maximum mass error is 9.16×10^{-5} in float32 and 1.14×10^{-13} in float64; the tested cross-sample output and gradient dependencies are zero. These tests cover random, constant, nearly constant and outlier inputs across $N \in \{4, 16, 64, 392\}$ and several budgets. This is a correctness reference, not a fused production kernel or a trained replacement for the reported model. Finite gradients can still be large near a zero score range. Appendix G subsequently tests GPU/autocast inference with local CIFAR weights. The later campaign in Section 5.3 trains a float32/Triton rowwise implementation, which is distinct from this slow float64 reference. Its completed CIFAR runs do not establish ImageNet training behavior. Mass tolerances in these checks are 5×10^{-4} for float32 and 10^{-6} for float64.

The artifact subset includes unmodified per-run summaries, resolved arguments with host paths removed, launch-protocol extracts, sanitized successful sweep records, audited benchmark tables, and source hashes. The CPU rebuild script recomputes best/final metrics, resolves sweep records by timestamp, checks numeric joins, and regenerates tables and figures. It never launches training. Source and claim ledgers retain their literature retrieval dates; the collaborator-packet audit was refreshed on September 21. Company policy prevents ImageNet weight transfer; independent ImageNet checkpoint re-evaluation is unavailable. The September 19 packet adds completed 512-pixel, curriculum, and hyperparameter runs. The refreshed audit matches 3,264 of 3,365 summary rows to delivered EMA logs, leaving 101 rows without corresponding log records (Section 4). Those rows are not discarded, but retain their summary-only evidence status.

D DeiT-B/16 is not a usable scale anchor

A 300-epoch DeiT-B/16 run at 224 pixels completes under the same global-batch and seed protocol as the small-scale campaign, with drop-path 0.1 inherited from the DeiT-S configurations. Table 20 reports

the peak and final EMA scores. Train loss continues to fall while evaluation loss rises after the peak at epoch 179. The final top-1, 78.566%, is below our DeiT-S/16 result of 79.692% at the same 196-token budget. Published DeiT-B 300-epoch training uses drop-path 0.2. We therefore do not append this run to the DeiT-S/8 table, and we do not launch a paired MergeNet-B comparison from these weights.

Epoch	Train loss	Eval loss	EMA top-1 (%)
150	2.968	0.889	80.060
179 (peak)	2.808	0.895	80.442
299 (final)	2.348	1.095	78.566

Table 20: DeiT-B/16, 300 epochs, drop-path 0.1. Peak and final scores must be reported together; the peak is not a converged result.

E A Bounded Layout Optimization Experiment

We additionally test a single LocalAttention module on an A100-SXM4-80GB. The original module constructs query/key/value tensors in batch-head-token-channel order, then transposes and copies them for the FlashAttention wrapper. An isolated variant constructs batch-token-head-channel tensors directly and adjusts the global class-query attention views and final reshape accordingly. It does not alter the released model, load a checkpoint, or apply an optimizer update.

The test uses PyTorch 2.6.0+cu124 and FlashAttention 2.7.4.post1, width 384, six heads, window radius 16, zero attention dropout, float32 parameters and inputs, and fp16 autocast. Four batch-2 parity cases cover sequence lengths 785 and 2,305, global class attention on/off, and query/key normalization. Outputs and input gradients are bitwise equal in these cases. Parameter gradients are also equal except for the normalization case, whose maximum relative L_2 error is 5.77×10^{-7} . This coverage does not establish all-shape or dropout/RNG-preserving equivalence for a complete model.

Batch	Tokens	Mode	Original ms	Layout ms	Reduction range
64	785	Forward	1.5600	1.0448	13.50–34.03%
64	785	Forward + backward	5.1590	4.3999	14.68–25.64%
16	2305	Forward	1.0318	0.9369	6.88–36.68%
16	2305	Forward + backward	5.3740	4.9176	5.45–11.83%

Table 21: Single-module A100 measurements. Times are medians of three round medians, each with five warmups and twenty CUDA-event samples. The final column is the range of paired-round time reductions, not a confidence interval. Tokens include the class token. Backward includes a squared-output loss and gradient clearing, with no optimizer.

Measurement order alternates by round, and the harness requires an initially empty device, establishes its CUDA process identity, then rejects a changed process set at subsequent checks. All iteration samples and snapshots are retained. The observed reductions support this local engineering direction, but the round variation is appreciable. Both module instances are resident, so their recorded memory peaks are not independent per-model memory measurements. These results do not predict the full model’s speedup, validate an HBM-traffic estimate, transfer H20 timings to A100, or resolve the separate ToMe attention-policy mismatch.

F Historical CIFAR-100 spatial-routing ablation

We retain a historical CIFAR-100 experiment as a limited diagnostic of the routing geometry. The August campaign contains 60 new runs: two MergeNet configurations, five input sizes, and six routing settings (Global and Euclidean radii $R \in \{1, 2, 3, 4, 8\}$), all using seed 42. The configurations have local/latent depths of 6/6 with final compression ratio $\lambda = 2$, or 4/8 with $\lambda = 4$; both use width 384, six heads, and 8×8 patches. Each model was trained from scratch for 200 epochs without distillation. Training used AdamW, learning rate 10^{-3} , weight decay 0.05, a cosine schedule with 20 warmup epochs, effective batch size 200, and EMA decay 0.9998. The primary endpoint is the final epoch-199 EMA top-1 accuracy on the 10,000-image CIFAR-100 test set. The frozen local campaign protocol designated $R = 3$ as the primary setting before the recorded training launches; this is an internal protocol record, not an external preregistration. We keep that setting even though $R = 2$ has a slightly higher mean.

The original images are 32×32 pixels and are resized to the listed input sizes. Thus, this sweep changes the patch grid over upsampled images and does not test native high-resolution image content. Evaluation used FP16 autocast and deterministic fast grouping, with batches of 200, 200, 200, 100, and 50 at sizes 160, 192, 224, 256, and 320, respectively. Global and the finite-radius settings were trained in the same frozen implementation and share the non-routing configuration. The additional Flat reference uses the historical flattened-window implementation with width parameter 8 and comes from ten earlier August runs. Its recorded non-routing training arguments and dependency versions match, but its runtime code snapshot differs; three Flat runs resumed within their original jobs. We therefore use Flat as a historical descriptive reference.

Configuration	Input	Global	Flat [†]	$R = 3$	ΔG	ΔF
6/6, $\lambda = 2$	160	67.80	68.79	69.42	+1.62	+0.63
	192	69.42	69.93	70.85	+1.43	+0.92
	224	69.50	70.53	70.53	+1.03	+0.00
	256	69.50	70.12	71.03	+1.53	+0.91
	320	69.85	70.81	71.42	+1.57	+0.61
4/8, $\lambda = 4$	160	65.29	66.69	67.92	+2.63	+1.23
	192	69.87	69.75	71.27	+1.40	+1.52
	224	68.21	69.87	70.52	+2.31	+0.65
	256	67.35	68.30	69.40	+2.05	+1.10
	320	66.05	67.08	68.83	+2.78	+1.75
Mean		68.284	69.187	70.119	+1.835	+0.932

Table 22: Historical CIFAR-100 final-epoch EMA accuracy (%) and $R = 3$ deltas (percentage points) against Global (ΔG) and Flat (ΔF). Each row uses seed 42; the mean gives equal weight to the ten configuration–input pairs. [†]Flat is a historical reference with a different code snapshot. Input denotes the resized side length of native 32×32 images. These are observations under the stated evaluation batches and historical dense-score implementation.

At the fixed $R = 3$ setting, accuracy exceeds Global in all ten pairs, with a mean improvement of 1.835 percentage points. Relative to historical Flat, nine pairs improve and one ties, with a mean difference of 0.932 points. Across the complete sweep, the mean accuracies for Global and radii 1, 2, 3, 4, and 8 are 68.284, 68.774, 70.225, 70.119, 70.047, and 68.414%, respectively. Radii 2, 3, and 4 each exceed their matched Global control in all ten pairs. These correlated, single-seed observations provide evidence that restricting routing neighborhoods can help in this particular setting; they do not establish statistical significance, native-resolution transfer, or superiority over external token-reduction methods.

The audit reconciled all 60 new endpoints and the ten Flat endpoints with raw 200-epoch summaries, completion receipts, and frozen manifests. Archived full-test evaluations of the 60 new epoch-199 EMA checkpoints show identical generic- and fast-grouping logits; their accuracies differ from the training summaries in 31 cases by at most 0.02 points. The table preserves the registered training-summary endpoint. The initial historical audit checked receipt hashes and checkpoint presence, without freshly evaluating or rehashing all checkpoints. The historical spatial implementation constructs a dense donor–receiver score matrix before masking, so these results do not validate the later sparse backend or its efficiency. Furthermore, both historical implementations use a threshold operator whose normalization range is shared across a batch, allowing a sample’s routing to depend on its batch companions. Appendix G subsequently rehashes and reevaluates two of these checkpoints and quantifies their batch sensitivity; the full historical sweep remains tied to its recorded evaluation batches. Section 5.3 reports a later 24-run campaign with

three seeds, two selectors, and a degree-matched control; those runs are independent trainings and are not substitutes for the table above.

G Independent Follow-up with Available Local Artifacts

Company policy prevents transfer of the ImageNet checkpoints. We therefore separate two feasible follow-ups: architecture timing using random weights, and prediction tests using existing, locally available CIFAR-100 weights. Neither experiment independently reproduces the company ImageNet scores.

G.1 A common A100 inference harness

We compare dense DeiT-S/8, ToMe, and the canonical sparse-backend MergeNet on an A100-SXM4-80GB using PyTorch 2.6.0+cu124, timm 0.9.11 and FlashAttention 2.7.4.post1. All cells use batch 64, float32 parameters and inputs, and fp16 autocast. The compressed models retain exactly 392 patches at 224 pixels and 1152 at 384 pixels, excluding CLS. A diagnostic forward records block outputs; all hooks are removed before timing. ToMe source tracing is disabled. Proportional attention is explicitly controlled, with its enabled setting matching the recorded accuracy protocol.

Each cell has ten warm-up iterations and thirty CUDA-event samples in each of three rounds, reversing model order in the middle round. A selected-GPU process guard requires an empty device before setup and only our process during the experiment. Models are allocated and released sequentially. Table 23 reports the median of three round medians. It measures model execution, excluding data loading, backward and updates.

Model	224 px (ms)	384 px (ms)
Dense DeiT-S/8	30.09	109.43
ToMe, proportional attention on	34.42	133.36
ToMe, proportional attention off	29.71	96.05
MergeNet $R = 3$	49.16	144.97
MergeNet $R = 3$, BNHD	46.56	138.11
MergeNet $R = 5.143$	–	218.25
MergeNet $R = 5.143$, BNHD	–	211.40

Table 23: Independent synthetic A100 inference, batch 64. All weights are randomly initialized; these are not trained-checkpoint frontier points. BNHD changes the local-attention tensor layout. The 384-pixel radius-5.143 model has the same final patch budget as radius 3.

On this local hardware/software stack, the original MergeNet is slower than dense DeiT at both resolutions. The BNHD change reduces complete-model radius-3 time by about 5%, considerably less than the isolated-module improvement. The proportional-attention setting materially affects ToMe latency. Thus neither the H20 crossover nor a ToMe timing measured with proportional attention disabled should be treated as accelerator-independent evidence. The A100 comparison changes the hardware and software stack relative to the H20 experiment; it does not isolate a hardware-only cause. No historical accuracy point is attached to these random-weight timings.

G.2 Layout validation under a complete training graph

The layout comparison uses identical weights and images, restored CPU and CUDA random states, fresh per-sample routing, and zero dropout at both 224 and 384 pixels. With the dense routing reference backend, outputs, input gradients and all parameter gradients are bitwise equal. With the sparse backend and an unscaled squared-logit loss, the strict gradient comparison fails; an unchanged-model repeat also fails, so these runs do not isolate an error introduced by BNHD. Their input-gradient relative errors are approximately 4.8% and 16.3%, respectively.

We retain these failed tests and add a fixed loss scale of 1024, unscaling all gradients before comparison. The sparse-backend tests then pass the predefined relative- L_2 limits of 0.002 for outputs and 0.003 for gradients. Maximum parameter-gradient relative errors are below 0.0016; unchanged-model repeat controls are below 0.0018. This diagnoses sensitivity of the unscaled fp16 test and supports bounded layout parity. It does not establish long-run training equivalence, eliminate sparse-kernel nondeterminism, or validate arbitrary dropout and precision settings.

G.3 Batch sensitivity in locally trained CIFAR models

We strictly load the epoch-199 EMA states for the $\lambda = 2$, 224-pixel global and radius-3 models from the August spatial campaign. Fresh file hashes match the completion receipts, and saved model arguments are checked against the frozen protocol. The original dense routing runtime and CIFAR normalization are retained. Every evaluation covers the same 10,000 test images. The batch-200 baseline reproduces radius-3 accuracy exactly and is 0.01 pp below the recorded global endpoint; it is not claimed to reproduce both endpoints bitwise.

Evaluation	Global (%)	$R = 3$ (%)	Gap (pp)	Changed G/R3
Original order, $B = 200$	69.49	70.53	+1.04	0/0
Original order, $B = 64$	69.41	70.55	+1.14	168/35
Permuted order, $B = 200$	69.44	70.52	+1.08	163/28
Rowwise reference, $B = 200$	69.17	70.55	+1.38	548/114

Table 24: Independent local CIFAR EMA evaluation. “Changed” counts the images whose predicted class differs from that model’s original-order batch-200 result. The last row replaces the historical selector by the rowwise bisection reference without updating weights. It is a distinct inference intervention, not the historical model’s score.

Changing batch size or image order changes 163–168 global-model predictions and 28–35 radius-3 predictions, while top-1 changes remain at most 0.08 pp and 0.02 pp, respectively. The radius-3 advantage remains 1.04–1.14 pp in these three protocols. Thus the implementation issue is observable in a trained model, but it does not erase the geometry benefit in this bounded pair of checkpoints.

A separate probe holds the first 16 images fixed in a batch of 64 and replaces only the other 48 images. Maximum logit changes are 0.335938 for global routing and 0.048828 for radius 3; exact same-batch repeats give zero change. Substituting the per-row bisection reference gives zero change in both probes. This intervention supports a selector-level explanation for these particular batch-composition effects. None of the first 16 predicted classes changes in this small probe; full-set class changes are reported separately above.

The full-set selector intervention changes 548 global predictions and 114 radius-3 predictions. Global top-1 falls by 0.32 pp, while radius-3 changes by +0.02 pp. Consequently, restoring per-sample selection is not a general accuracy-preserving substitution for old weights. The reference passes this bounded GPU inference check, but remains an untrained correctness implementation. This checkpoint intervention alone does not assess its learning behavior; the subsequent training experiment is described below.

A subsequent 24-run CIFAR-100 campaign trains both selectors from scratch with three seeds and a degree-matched control (Section 5.3). Those endpoints are new trainings, not re-evaluations of the two August checkpoints in this appendix.

The artifact includes all per-image predicted classes, labels, raw timing samples, observed token shapes and hash receipts. These local tests address specific implementation and comparison questions; two seed-42 CIFAR models cannot establish multi-seed reliability or ImageNet performance after a selector change.

H Reserved Extensions and Outstanding Evidence

These are intentionally unfilled working-draft slots, not additional results. Optional experiments may be omitted together with the associated claims. Original observations and partial runs remain in the evidence inventory.

H.1 Base-scale comparison

Evidence pending [P09]: Establish a credible dense reference before a scale claim

The completed drop-path-0.1 DeiT-B run remains a reported negative observation.

Specify matched Base-scale training and effective layerwise stochastic depth. Diagnose the dense reference’s overfitting before interpreting a MergeNet-B comparison. A different regularization recipe is a new experiment, not a correction that replaces the old result.

Arm	Seeds	Best EMA	Final EMA	Batch latency (ms)
New dense Base reference	_____	_____	_____	_____
Matched MergeNet-B	_____	_____	_____	_____

Result and interpretation:

H.2 Same-parent radius comparison at 512 pixels

Evidence pending [P10]: Separate radius from checkpoint ancestry

Optional; the existing 512-pixel result is valid for its recorded radius-5.143 parent.

A radius-only inference comparison uses one checkpoint with other settings fixed. A fine-tuning comparison initializes both radii from the same verified 384-pixel state and uses the same 15-epoch recipe. An EMA extraction from the radius-3 checkpoint permits that branch, but comparing a new R3-parent branch only to the old R5.143-parent branch still mixes parent and radius.

512-pixel arm	Parent identity	Best EMA	Final EMA
Fixed radius	_____	_____	_____
Scaled radius, same parent	_____	_____	_____

Result and interpretation:

H.3 Weight-decay probe completion

Evidence pending [P11]: Complete only if a full weight-decay result is needed

The 140/150 snapshot is neither a completed negative result nor a statistical null.

If resumed, retain the original state and append epochs 140–149 without duplicating earlier rows. Otherwise keep partial status and omit endpoint claims. Not crossing a decision threshold is not statistical equivalence.

Run	Completed epochs	Best EMA	Final EMA
Weight decay 0.03	_____	_____	_____

Result and interpretation:

H.4 Additional log and state receipts

Evidence pending [P12]: Complete source-to-result linkage without transferring weights

Existing configurations and environment records need not be re-exported unless actual execution differs.

For each new result provide run ID, complete metrics, resolved arguments including launcher overrides, startup/resume lines, actual code-file hashes and environment versions. Evaluations and initialization also need checkpoint digest, state key, epoch, EMA flag and tensor mapping or verification receipt. Digests identify artifacts without sharing parameter values. If a receipt cannot be disclosed, record that evidence boundary.

Some September 19 summaries extend past the copied logs. Missing segments must be supplied or labeled summary-only. The companion audit identifies exact gaps and excludes documented earlier attempts; CSV recalculation is not checkpoint replay.

Additional receipt	Expected epochs / key	Verified count	Digest
Curriculum missing EMA logs	64–149	_____	_____
MergeNet 512-pixel EMA logs	0–14	_____	_____
Initialization state / full mapping	raw / EMA	_____	_____
New run source manifest	per run	_____	_____

Result and interpretation:
